CS 6140: Machine Learning

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Grading

- **Homeworks** (40% of the grade)
  - 4 to 5 HWs: analytical + programming assignments (Python, Pytorch)
  - **No late** HWs, submit on **Canvas** by specified deadline, done **individually**
- **Midterm 1 and 2** (30% of the grade, each 15%)
  - **Feb 27 and April 17**, 1 cheat sheet (a4 or letter size, both sides)
- **Project** (30% of the grade)
  - **Project teams**: **Feb 27**; teams of 1 to 3 people
  - **Project pitch + proposal**: **March 10**; pick project, short presentation
  - **Project report**: **April 24** (format will be posted on piazza)
- **Class participation** (up to +5%)

Grading

- Class Project
  - Teams of **1 to 3** people
  - **Same evaluation** criteria
  - **Encourage** working in teams of 2 or 3
  - Submit names of your team members by **Feb 27th, 11:59pm** on canvas

- List of **potential topics**
  - Audio & music
  - Computer vision
  - Robotics
  - Finance & Commerce
  - Natural Language
  - Physical Sciences
  - Sports & Athletics
  - Life Sciences
  - Education

Project Grade Breakdown

- **Project Presentation/Pitch**: **15%**
  - Duration: ~3min per team (exact duration TBA)
- **Project Proposal**: **15%**
  - Template .tex/.doc files uploaded on piazza/General resources
- **Final Project Report**: **50%**
  - Sample project report uploaded on piazza/General resources
- **Project Code**: **20%**
  - Share GitHub link to your well-documented code (we should be able to follow installation guidelines and replicate results)

Project (Expectations)

- Pick a **real-world problem** you are interested in (care about)
- A **real dataset** should be available (free) to work with
  - **Sufficiently large** (> ~500 for each class)

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  - >= 3 classifiers: e.g., logistic, SVM, DNNs, … [**Must have: DNNs**]
  - >= 3 regressors: e.g., Linear regression with different basis function expansions, Random Forest and DNNs [**Must have: DNNs**]

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- Apply the methods to data and **report results**
  - Investigate performance comparison of different methods
  - Investigate effect of hyper-parameters (cross-validation)

Toolboxes

- What **toolboxes** can I use?
  - **Scikitlearn** is acceptable for implementation of logistic, SVM, etc
  - For Neural Networks, use **Pytorch**

Toolboxes

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  - For Neural Networks, use **Pytorch**

- Can I use R, Matlab, etc?
  - **No** in general. But, you can discuss with me…

Proposal

- **Project proposal**: **March 10** **(will hold the lecture over Zoom)**
- You **may receive feedback to modify**
  - e.g., I might suggest to include classifier X, or compare with Y
  - If you do not receive, that means all good :-)

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- Proposal **templates** available on piazza/resources
  - **Two pages** (latex/word): **abstract**, **introduction/motivation**, **datasets** that will be used (give enough information: where it comes from, how many classes, how many samples in each class, what kind of features will be used, what preprocessing will be applied on data,…)
  - Two pages excluding references and figures

(Potential) Project Ideas

- Stanford CS229: http://cs229.stanford.edu/projects.html

Not Secure | cs229.stanford.edu/projects.html

CS229 Announcements Syllabus Course Info Logistics Projects Piazza

CS229 Final Project Information

One of CS229's main goals is to prepare you to apply machine learning algorithms to real-world tasks, or to leave you well-qualified to start machine learning or AI research. The final project is intended to start you in these directions.

For group-specific questions regarding projects, please create a private post on Piazza. Please first have a look through the frequently asked questions. Note: only one group member is supposed to submit the assignment, and tag the rest of the group members (do not all submit separately, or on the flip side forget to tag your teammates if you are the group's designated submitter). If you do not do this, you can submit a regrade request and we will fix it, but we will also deduct 1 point.

Previous projects

2019 (Autumn)	2019 (Spring)	2018	2017	2016	2016 (Spring)	2015
2014	2013	2012	2011	2010	2009	
2008	2007	2006	2005	2004		

Project Topics

Your first task is to pick a project topic. If you're looking for project ideas, please come to project office hours, and we'd be happy to brainstorm and suggest some project ideas. In the meantime, here are some suggestions that might also help.

Most students do one of three kinds of projects:

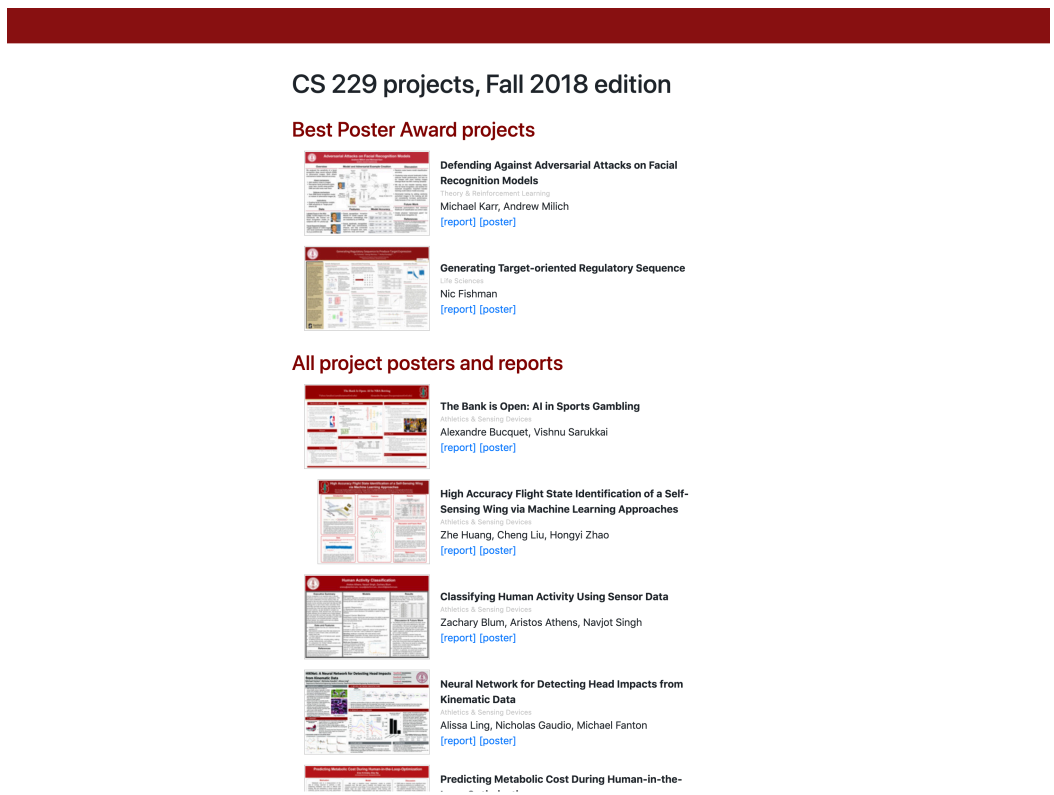
1. **Application project**. This is by far the most common: Pick an application that interests you, and explore how best to apply learning algorithms to solve it.
2. **Algorithmic project**. Pick a problem or family of problems, and develop a new learning algorithm, or a novel variant of an existing algorithm, to solve it.
3. **Theoretical project**. Prove some interesting/non-trivial properties of a new or an existing learning algorithm. (This is often quite difficult, and so very few, if any, projects will be purely theoretical.)

Some projects will also combine elements of applications, algorithms and theory.

Many fantastic class projects come from students picking either an application area that they're interested in, or picking some subfield of machine learning that they want to explore more. So, pick something that you can get excited and passionate about! Be brave rather than timid, and do feel free to propose ambitious things that you're excited about. (Just be sure to ask us for help if you're uncertain how to best get started.) Alternatively, if you're already working on a research or industry project that machine learning might apply to, then you may already have a great project idea.

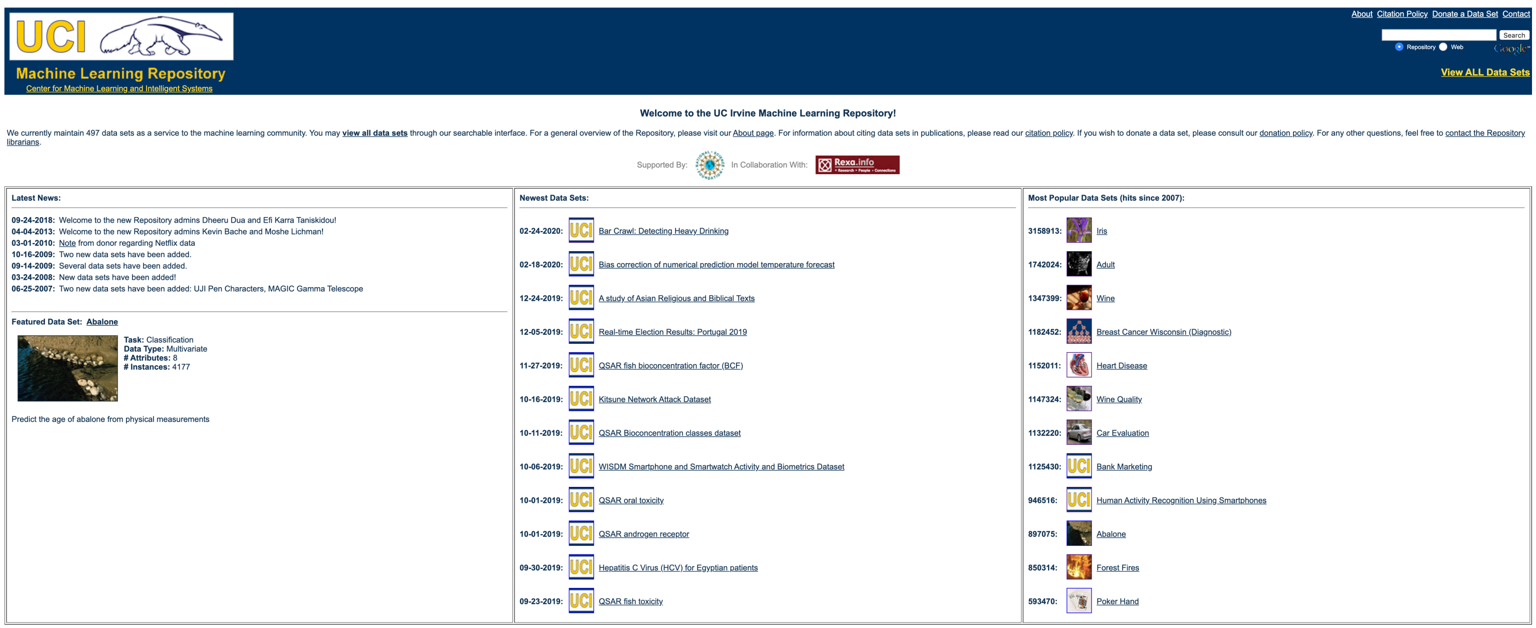
(Potential) Project Ideas

- Stanford CS229: http://cs229.stanford.edu/projects.html
  - Look into posters/report: many have dataset info and good gist of ideas



(Potential) Project Resources

- UCI ML repository



(Potential) Project Resources

- Click on **view all datasets**: see page below
  - Look under area for categories of your interest

UCI
Machine Learning Repository
Center for Machine Learning and Intelligent Systems

Browse Through: **497** Data Sets

Table View List View

Default Task
Classification (366)
Regression (112)
Clustering (92)
Other (55)
Attribute Type
Categorical (38)
Numerical (331)
Mixed (55)
Data Type
Multivariate (383)
Univariate (24)
Sequential (52)
Time-Series (102)
Text (57)
Domain-Theory (23)
Other (21)
Area
Life Sciences (113)
Physical Sciences (55)
CS / Engineering (179)
Social Sciences (28)
Business (32)
Game (10)
Other (75)
 # Attributes
Less than 10 (122)
10 to 100 (223)
Greater than 100 (90)
 # Instances
Less than 100 (27)
100 to 1000 (169)
Greater than 1000 (267)
Format Type
Matrix (347)
Non-Matrix (150)

Name	Data Types	Default Task	Attribute Types	# Instances	# Attributes	Year
Abalone	Multivariate	Classification	Categorical, Integer, Real	4177	8	1995
Adult	Multivariate	Classification	Categorical, Integer	48842	14	1996
Annealing	Multivariate	Classification	Categorical, Integer, Real	798	38	
Anonymous Microsoft Web Data		Recommender-Systems	Categorical	37711	294	1998
Arrhythmia	Multivariate	Classification	Categorical, Integer, Real	452	279	1998
Artificial Characters	Multivariate	Classification	Categorical, Integer, Real	6000	7	1992
Audiology (Original)	Multivariate	Classification	Categorical	226		1987
Audiology (Standardized)	Multivariate	Classification	Categorical	226	69	1992
Auto MPG	Multivariate	Regression	Categorical, Real	398	8	1993
Automobile	Multivariate	Regression	Categorical, Integer, Real	205	26	1987
Badges	Univariate, Text	Classification		294	1	1994
Balance Scale	Multivariate	Classification	Categorical	625	4	1994

(Potential) Project Resources

- E.g., for CS/Engineering
  - Select ones where dataset type: classification, or regression

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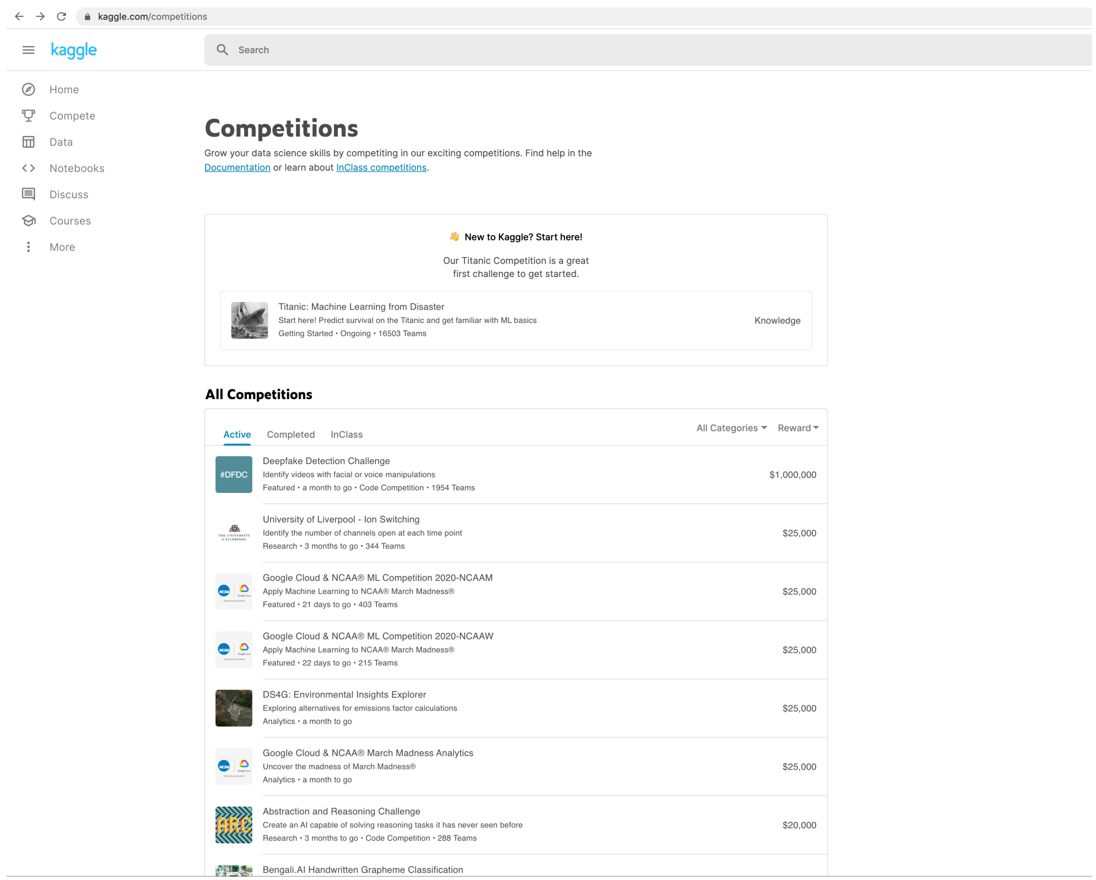
(Potential) Project Resources

- Kaggle competitions



Sample Projects

- **Heart Disease Recognition**: Distinguish presence (values 1,2,3,4) from absence (value 0)

Heart Disease Data Set

Download: Data Folder, Data Set Description

Abstract: 4 databases: Cleveland, Hungary, Switzerland, and the VA Long Beach

Data Set Characteristics:	Multivariate	**Number of Instances:**	303	**Area:**	Life
Attribute Characteristics:	Categorical, Integer, Real	**Number of Attributes:**	75	**Date Donated**	1988-07-01
Associated Tasks:	Classification	**Missing Values?**	Yes	**Number of Web Hits:**	1151992

Attribute Information:

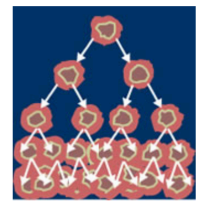
Only 14 attributes used:
1. #3 (age)
2. #4 (sex)
3. #9 (cp)
4. #10 (trestbps)
5. #12 (chol)
6. #16 (fbs)
7. #19 (restecg)
8. #32 (thalach)
9. #38 (exang)
10. #40 (oldpeak)
11. #41 (slope)
12. #44 (ca)
13. #51 (thal)
14. #58 (num) (the predicted attribute)

Sample Projects

- **Cancer classification** using digitized image of a fine needle aspirate of breast mass (describe characteristics of the cell nuclei)

Breast Cancer Wisconsin (Diagnostic) Data Set
Download: Data Folder, Data Set Description

Abstract: Diagnostic Wisconsin Breast Cancer Database



Data Set Characteristics:	Multivariate	**Number of Instances:**	569	**Area:**	Life
Attribute Characteristics:	Real	**Number of Attributes:**	32	**Date Donated**	1995-11-01
Associated Tasks:	Classification	**Missing Values?**	No	**Number of Web Hits:**	1182421

Attribute Information:

1) ID number
2) Diagnosis (M = malignant, B = benign)
3-32)

Ten real-valued features are computed for each cell nucleus:

a) radius (mean of distances from center to points on the perimeter)
b) texture (standard deviation of gray-scale values)
c) perimeter
d) area
e) smoothness (local variation in radius lengths)
f) compactness (perimeter^2 / area - 1.0)
g) concavity (severity of concave portions of the contour)
h) concave points (number of concave portions of the contour)
i) symmetry
j) fractal dimension ("coastline approximation" - 1)

Sample Projects

- Classify the **origin of wines** using chemical analysis

Wine Data Set

Download: Data Folder, Data Set Description

Abstract: Using chemical analysis determine the origin of wines

Data Set Characteristics:	Multivariate	**Number of Instances:**	178	**Area:**	Physical
Attribute Characteristics:	Integer, Real	**Number of Attributes:**	13	**Date Donated**	1991-07-01
Associated Tasks:	Classification	**Missing Values?**	No	**Number of Web Hits:**	1347380

Source:

Original Owners:

Forina, M. et al, PARVUS -
An Extendible Package for Data Exploration, Classification and Correlation.
Institute of Pharmaceutical and Food Analysis and Technologies, Via Brigata Salerno,
16147 Genoa, Italy.

Donor:

Stefan Aeberhard, email: stefan '@' coral.cs.jcu.edu.au

Data Set Information:

These data are the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. The analysis determined the quantities of 13 constituents found in each of the three types of wines.

I think that the initial data set had around 30 variables, but for some reason I only have the 13 dimensional version. I had a list of what the 30 or so variables were, but a.) I lost it, and b.), I would not know which 13 variables are includ

The attributes are (dontated by Riccardo Leardi, riclea '@' anchem.unige.it)
1) Alcohol
2) Malic acid
3) Ash
4) Alcalinity of ash
5) Magnesium
6) Total phenols
7) Flavanoids
8) Nonflavanoid phenols
9) Proanthocyanins
10)Color intensity
11)Hue
12)OD280/OD315 of diluted wines
13)Proline

Sample Projects

- **Human action recognition** from smart phone data

Human Activity Recognition Using Smartphones Data Set

Download: Data Folder, Data Set Description

Abstract: Human Activity Recognition database built from the recordings of 30 subjects performing activities of daily living (ADL) while carrying a waist-mounted smartphone with embedded inertial sensors.

Data Set Characteristics:	Multivariate, Time-Series	**Number of Instances:**	10299	**Area:**	Computer
Attribute Characteristics:	N/A	**Number of Attributes:**	561	**Date Donated**	2012-12-10
Associated Tasks:	Classification, Clustering	**Missing Values?**	N/A	**Number of Web Hits:**	946504

Source:

Jorge L. Reyes-Ortiz(1,2), Davide Anguita(1), Alessandro Ghio(1), Luca Oneto(1) and Xavier Parra(2)
1 - Smartlab - Non-Linear Complex Systems Laboratory
DITEN - Università degli Studi di Genova, Genoa (I-16145), Italy.
2 - CETpD - Technical Research Centre for Dependency Care and Autonomous Living
Universitat Politècnica de Catalunya (BarcelonaTech). Vilanova i la Geltrú (08800), Spain
activityrecognition '@' smartlab.ws

Data Set Information:

The experiments have been carried out with a group of 30 volunteers within an age bracket of 19-48 years. Each person performed six activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, L
constant rate of 50Hz. The experiments have been video-recorded to label the data manually. The obtained dataset has been randomly partitioned into two sets, where 70% of the volunteers was selected for generating the training dat

The sensor signals (accelerometer and gyroscope) were pre-processed by applying noise filters and then sampled in fixed-width sliding windows of 2.56 sec and 50% overlap (128 readings/window). The sensor acceleration signal, whi
frequency components, therefore a filter with 0.3 Hz cutoff frequency was used. From each window, a vector of features was obtained by calculating variables from the time and frequency domain.

Check the README.txt file for further details about this dataset.

A video of the experiment including an example of the 6 recorded activities with one of the participants can be seen in the following link: [Web Link]

An updated version of this dataset can be found at [Web Link]. It includes labels of postural transitions between activities and also the full raw inertial signals instead of the ones pre-processed into windows.

Attribute Information:

For each record in the dataset it is provided:
- Triaxial acceleration from the accelerometer (total acceleration) and the estimated body acceleration.

Sample Projects

- Predict **burned area of forest** using meteorological data

Forest Fires Data Set

Download: Data Folder, Data Set Description

Abstract: This is a difficult regression task, where the aim is to predict the burned area of forest fires, in the northeast region of Portugal, by using meteorological and other data (see details at: [Web Link]).

Data Set Characteristics:	Multivariate	**Number of Instances:**	517	**Area:**	Physical
Attribute Characteristics:	Real	**Number of Attributes:**	13	**Date Donated**	2008-02-29
Associated Tasks:	Regression	**Missing Values?**	N/A	**Number of Web Hits:**	850300

Source:

Paulo Cortez, pcortez '@' dsi.uminho.pt, Department of Information Systems, University of Minho, Portugal.
Aníbal Morais, araimorais '@' gmail.com, Department of Information Systems, University of Minho, Portugal.

Data Set Information:

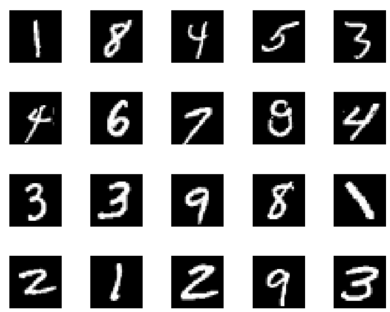
In [Cortez and Morais, 2007], the output 'area' was first transformed with a ln(x+1) function. Then, several Data Mining methods were applied. After fitting the models, the outputs were post-processed with the inverse of the ln(x+1) transform. Four different input setups were used. The experiments were conducted using a 10-fold (cross-validation) x 30 runs. Two regression metrics were measured: MAD and RMSE. A Gaussian support vector machine (SVM) fed with only 4 direct weather conditions (temp, RH, wind and rain) obtained the best MAD value: 12.71 +- 0.01 (mean and confidence interval within 95% using a t-student distribution). The best RMSE was attained by the naive mean predictor. An analysis to the regression error curve (REC) shows that the SVM model predicts more examples within a lower admitted error. In effect, the SVM model predicts better small fires, which are the majority.

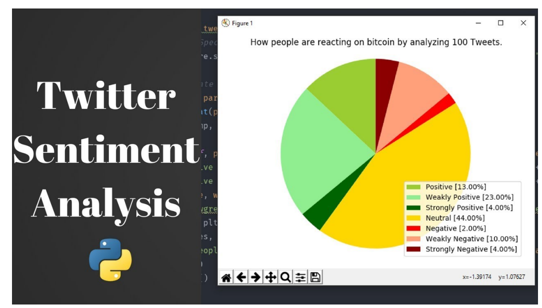
Attribute Information:

For more information, read [Cortez and Morais, 2007].
1. X - x-axis spatial coordinate within the Montesinho park map: 1 to 9
2. Y - y-axis spatial coordinate within the Montesinho park map: 2 to 9
3. month - month of the year: 'jan' to 'dec'
4. day - day of the week: 'mon' to 'sun'
5. FFMC - FFMC index from the FWI system: 18.7 to 96.20
6. DMC - DMC index from the FWI system: 1.1 to 291.3
7. DC - DC index from the FWI system: 7.9 to 860.6
8. ISI - ISI index from the FWI system: 0.0 to 56.10
9. temp - temperature in Celsius degrees: 2.2 to 33.30

Sample Projects

- Other projects:
  - Music Genre Classification (rock, pop, jazz, etc)
  - Oil/stock price modeling and prediction
  - Image/digit/text classification
  - Sentiment classification from tweet/news
  - …





Past Projects

Human vs AI Generated Image Detection

Abstract

The growing realism of AI-generated images poses significant challenges to media authenticity and public trust. As synthetic content becomes increasingly difficult to detect with the human eye, the need for automated detection systems has become critical. This project explores a binary classification approach to distinguish between real and AI-generated images using deep learning techniques. We evaluate a range of architectures, including Convolutional Neural Networks (CNNs), Vision Transformers (ViT), and ensemble methods, emphasizing their ability to capture both local features and global patterns. Through systematic experimentation and architectural comparisons, the project aims to contribute to the development of reliable tools for identifying synthetic media and mitigating the spread of misinformation.

(a) AI-Generated Image 1

(b) Real Image 1

In-Context Soccer Player Detection

Abstract

This project investigates state of the art models tasked with player detection in soccer (also known as football). Our project fine-tunes two models, a Faster-RCNN-ResNet50 and a YOLOv11n model, with context to analyze the improvements in performance. Beyond this, we also deep dive into the construction of a full scale Faster-RCNN. Our main goal is to analyze the baseline performance of the two models against their fine-tuned counterparts. We observe the effects of contextual tuning and analyze our methods of training and our results. Our secondary goal is exploring the architecture behind faster-rcnn's by stepping through the building process. Overall, the purpose of this project will be to provide analysis on our context focused method of fine-tuning models and how it can allow the models to improve. Leveraging advancements in deep learning, our methods aim to contribute to on-going research surrounding the limitations of modern VAR systems that depend on manual intervention and still-frame analyses.

Past Projects

Gesture Classification and Projection of EMG Data

Abstract

The primary focus of this project is to apply standard classification models such as support vector machines (SVM), random forests (RF), and deep neural networks (DNN) to instantaneous EMG sensor data to predict the associated hand gesture being performed. We then investigated the possibility of using transfer learning to "translate" samples from one EMG setup to another, allowing for the generation of new data that are consistent with the preferred EMG setup. Finally, we will discuss the importance of temporal context in processing EMG data and propose modifications that would allow this approach to make use of that valuable information.

Keywords Random Forest · SVM · Neural Networks · EMG · Transfer Learning

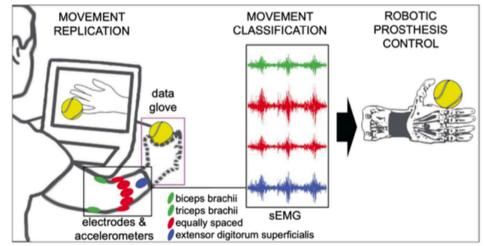
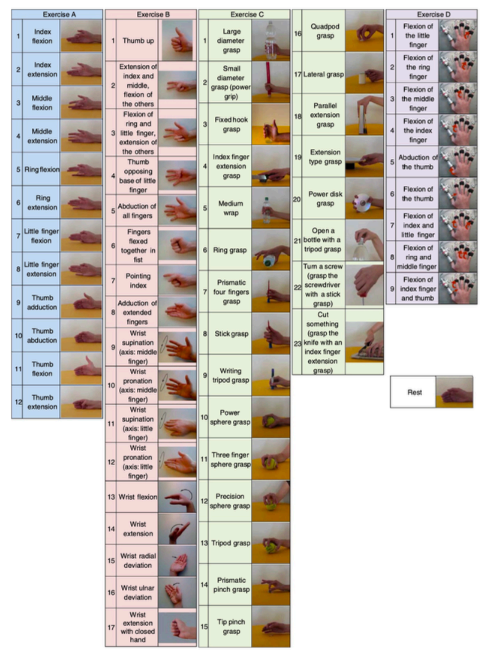


Figure 1: Ninapro Data Acquisition Protocol



Past Projects

Sentiment Analysis of IMDb Movie Reviews

[redacted]

Abstract

Sentiment Analysis refers to a technique that utilizes Natural Language Processing techniques in order to determine whether a statement is positive or negative. In this project, we will utilize Sentiment Analysis on IMDb movie reviews in order to judge whether a movie has a generally positive or negative favor among public perception. Employing different classification models (SVM, Random Forest, XG-Boost, LSTM, and BERT), we hope to compare their benchmarks to one another to determine which models are best for Sentiment Analysis in order to see which model would be best to use to determine whether the general sentiment about a movie is positive or negative.

Predicting Online Shoppers' Purchasing Intention

[redacted]

Abstract

We predict online shoppers' purchase intent using the *Online Shoppers Purchasing Intention* dataset. After one-hot encoding, standardisation, and SMOTE oversampling, five models—Logistic Regression, SVM, Random Forest, XGBoost, and a feed-forward DNN—are tuned with 5-fold stratified CV. XGBoost yields the best results (AUC = 0.926, $F_1 = 0.657$) while maintaining a small train–test gap; SMOTE raises minority-class recall across all methods. The study shows that gradient-boosted trees plus basic imbalance handling can deliver real-time intent scores to support targeted coupons and dynamic recommendations.

Predicting Next Day Wildfire Spread

[redacted]

Abstract

Wildfire spread prediction is a crucial task in environmental modeling, with significant implications for public safety, emergency planning, and ecological conservation. In this project, we investigate the Next Day Wildfire Spread dataset—a large-scale, remote-sensing dataset curated for day-ahead prediction of wildfire expansion. Our objective is twofold: (1) to replicate the baseline models proposed in the original benchmark study, and (2) to conduct a comparative analysis of classical and deep learning approaches under varying hyperparameter settings. Specifically, we implement and evaluate five models—Logistic Regression, Random Forest, Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and a Convolutional Autoencoder—across multiple learning rates and batch sizes. Our findings provide insight into the sensitivity of these models to training configurations and underscore the potential of deep architectures, particularly convolutional and recurrent models, in capturing complex spatial-temporal wildfire dynamics.